



Model Exam Ministry of Ethiopia

Information Technology

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All Questions with Answers
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1. In ER diagrammatic notation, has its name underlined inside the oval describes

- a. Complex attribute
- b. Multi-valued Attribute
- c. Key attribute
- d. Derived attribute

The correct answer is c. Key attribute.

In ER diagrammatic notation, the name of an entity's key attribute is usually underlined inside the oval that represents the entity. The key attribute is an attribute that uniquely identifies each

instance of an entity and is used to establish relationships between entities. It is also known as a primary key or identifier.

A complex attribute is an attribute that can be further divided into smaller sub-attributes, such as a person's address, which can be broken down into street, city, state, and zip code.

A multi-valued attribute is an attribute that can have multiple values for a single instance of an entity, such as a person's phone number or email address.

A derived attribute is an attribute that can be calculated or derived from other attributes of an entity, such as a person's age, which can be calculated from their date of birth.

2. Which of the following approaches is best for state decision support system software project management activities?

- a. Project execution
- b. Activity-based approach
- c. Hybrid approach
- d. Product-based approach

The correct answer is b. Activity-based approach.

The best approach for state decision support system (DSS) software project management activities depends on several factors, such as project scope, complexity, and organizational culture. However, out of the given options, the most appropriate approach for managing a DSS software project is the b. Activity-based approach.

The activity-based approach focuses on breaking down the project into smaller, manageable activities and tasks, and then assigning resources and timelines to each activity. This approach allows for greater control and visibility over the project's progress, as well as better coordination between team members and stakeholders.

The hybrid approach combines different project management methodologies and techniques to suit the specific needs of the project. This approach may be appropriate if the DSS software project involves a mix of technical and business requirements, or if the project team has diverse skill sets and expertise.

The project execution approach focuses on the efficient and effective implementation of the project plan, but it may not provide enough visibility and control over the project's progress or address potential risks and issues adequately.

The product-based approach focuses on the end product and its quality, but it may not provide enough guidance and direction for managing the project's activities and tasks.

In summary, the activity-based approach is the most appropriate approach for managing a state DSS software project's activities, as it provides greater control, visibility, and coordination over the project's progress and team members.

3. If you want to display an image without any text around it, you should nest it inside tag(s)?

a. `<p:img>`

b. `<img / p >`

c. `<img ><p>`

d. `<p>`

The correct option is b. `<img / >`.

To display an image without any text around it, you should nest it inside the tag, which is used specifically for embedding images in HTML documents. The tag is an empty element, which means it does not have a closing tag.

Here is an example of how to use the tag to display an image without any text around it:

```

```

In the example above, "image.jpg" is the path to the image file, and "Image description" is the text that will be displayed if the image cannot be loaded or accessed by the user.

Option a. <p:img> and option c. <p> are not valid HTML syntax.

Option d. <p> is a paragraph tag used to define a paragraph of text, and it is not appropriate for displaying images without text around them.

4. Routing loops are not a problem for link-state protocols like distance- vector protocols. Why?

- a. Link-state protocols share the topology database between all routers
- b. Link-state protocols require routers to maintain their own topology database of the network
- c. Link-state protocols use multiple routes to the same destination
- d. Link-state protocols allow routers to maintain a link- state database of all routers

The statement in the question is incorrect. Routing loops can be a problem for both link-state and distance-vector routing protocols.

Routing loops occur when packets are continuously routed between two or more routers in a network without reaching their destination. This problem can occur in both distance-vector and link-state protocols if proper routing protocols are not implemented.

To mitigate routing loops, both distance-vector and link-state protocols use various techniques, such as split horizon, poison reverse, and hold-down timers, to prevent loops from occurring.

To answer the given question, let's look at the options:

- a. Link-state protocols share the topology database between all routers - This statement is correct, but it does not explain why routing loops are not a problem for link-state protocols.
- b. Link-state protocols require routers to maintain their own topology database of the network - This statement is also correct, but it does not explain why routing loops are not a problem for link-state protocols.
- c. Link-state protocols use multiple routes to the same destination - This statement is incorrect. Link-state protocols and distance-vector protocols both support multiple routes to the same destination, but this does not prevent routing loops from occurring.
- d. Link-state protocols allow routers to maintain a link-state database of all routers - This statement is correct, but it does not explain why routing loops are not a problem for link-state protocols.

5. Is a database schema that depicts key dependencies between the primary key and foreign key.

- a. ER Diagram
- b. Database Diagram
- c. Relation Schema
- d. Schema Diagram

The correct answer is a. ER Diagram.

An ER (Entity-Relationship) diagram is a database schema that depicts key dependencies between the primary key and foreign key. It is a graphical representation of entities, their attributes, and the relationships among them.

In an ER diagram, entities are represented as rectangles, attributes are represented as ovals, and relationships are represented as diamonds. The lines connecting the entities and attributes represent the relationships between them. The primary key of an entity is typically underlined, and the foreign keys are represented by arrows pointing to the related entity.

A database diagram is a visual representation of a database that shows the tables, columns, relationships, and other database objects. It can include ER diagrams, as well as other types of diagrams, such as UML diagrams or flowcharts.

A relation schema is a formal description of the structure of a database that defines the tables, columns, and relationships among them. It is typically expressed in a formal language, such as SQL (Structured Query Language).

A schema diagram is a visualization of a database schema that shows the tables, columns, and relationships among them. It is similar to a database diagram but focuses specifically on the schema structure.

6. About the parts of a router, which of the following statements is true?

- a. A backup version of the IOS utilized during the boot process is kept in NVRAM
- b. ROM contains diagnostics that are run on the hardware modules
- c. The most recent and updated configuration files are located in the ROM
- d. A configuration file used during the boot process is permanently stored in RAM

The correct answer is b. ROM contains diagnostics that are run on the hardware modules.

Routers consist of various components that perform different functions. Here is a brief overview of the functions and characteristics of the components mentioned in the options:

a. NVRAM (Non-Volatile Random Access Memory) - This is a type of memory that is used to store the router's startup configuration file. It is non-volatile, which means that it retains its contents even when the router is turned off. The backup version of the IOS (Internetwork Operating System) is typically stored in a separate location, such as a flash memory card.

b. ROM (Read-Only Memory) - This is a type of memory that contains the router's bootstrap program, which is used to initialize the hardware and load the IOS software. ROM also contains diagnostics that are run on the hardware modules during the boot process to ensure that they are functioning correctly.

c. Configuration files - These are files that contain the settings and parameters that configure the operation of the router. The most recent and updated configuration files are typically stored in NVRAM or on a TFTP (Trivial File Transfer Protocol) server.

d. RAM (Random Access Memory) - This is a type of memory that is used to hold the IOS software, as well as the running configuration file. The configuration file used during the boot process is typically loaded from NVRAM or a TFTP server and stored in RAM.

In summary, option b is the correct statement as ROM contains diagnostics that are run on the hardware modules during the boot process.

7. Which one of the following is true about carrier Sense Multiple Access/Collision Detection (CSMA/CD) process on Ethernet LAN switch that configured with full-duplex?

- a. If two devices transmit at the same time, a collision will occur
- b. Data sent by both devices will not be corrupted and will need to be resent
- c. Devices will not detect the collision on the network
- d. Collision never occur the two devices transmit at the same time

The correct answer is d. Collision never occur the two devices transmit at the same time.

Carrier Sense Multiple Access/Collision Detection (CSMA/CD) is a protocol used in Ethernet networks to avoid collisions between devices attempting to transmit data over the network. In CSMA/CD, devices listen to the network to check if it is clear before attempting to transmit. If two devices try to transmit at the same time, a collision occurs, and both devices must stop transmitting, wait for a random amount of time, and then attempt to transmit again.

However, when an Ethernet LAN switch is configured with full-duplex, collisions cannot occur. In full-duplex mode, each device has a dedicated, point-to-point connection to the switch, which means that each device can transmit and receive data simultaneously without the risk of collisions. This is because the switch allocates a separate collision domain for each connected device, which eliminates the possibility of collisions.

Therefore, option d is correct - collisions do not occur when two devices transmit at the same time in a full-duplex Ethernet LAN switch.

8. What command is used to show the amount of space that a certain directory takes up on the disk?

- a. ds
- b. dd
- c. du
- d. dds

The correct answer is c. du.

The "du" command is used in Unix-based operating systems (such as Linux and macOS) to display the disk usage of a particular directory or file. It shows the amount of space that a directory or file takes up on the disk.

Here is the basic syntax of the "du" command:

```
du [options] [directory/file]
```

Some commonly used options with the "du" command are:

"-h": displays the output in a human-readable format

"-s": displays only a summary of the disk usage for the specified directory or file

"-c": displays a grand total of the disk usage for all directories or files

For example, to show the disk usage of the "/home/user/Documents" directory in a human-readable format, you can use the following command:

```
du -h /home/user/Documents
```

This will display the disk usage of the "/home/user/Documents" directory and all its subdirectories in a human-readable format.

9. Assume: Dawit created some nested tags as displayed here: `<p><i> Peace!
</i></p>`. Did he perform valid nesting? (Why?)

- a. No, because italic tags must always come before bold tags
- b. No, though the first part is right, the second part should be `</p></b></i>`
- c. No, because paragraph tags need to be right before the actual text
- d. Yes

The correct answer is d. Yes.

Yes, Dawit performed valid nesting with the given tags. The correct order of nesting tags in the example is: `<p>` (paragraph tag) -> `<b>` (bold tag) -> `<i>` (italic tag) -> "Peace!" (text) -> `</i>` (closing italic tag) -> `</b>` (closing bold tag) -> `</p>` (closing paragraph tag).

In HTML, it is possible to nest tags within one another to create a hierarchical structure. The order in which the tags are nested is important, and the innermost tag should always be closed first, followed by the outer tags.

In the given example, the text "Peace!" is enclosed within the `<i>` tag, which is enclosed within the `<b>` tag, and both of them are enclosed within the `<p>` tag. The closing tags are also in the correct order, with the `</i>` tag being closed first, followed by the `</b>` tag, and then the `</p>` tag.

Therefore, the given nesting of tags is valid and will render correctly in a web browser. Option d. "Yes, the nesting is valid" would have been the correct answer.

10. In case of any shut down during transaction before commit, which of the following statement is done automatically?

- a. Flashback
- b. Commit
- c. View
- d. Rollback

The correct answer is **d**. Rollback.

Rollback is a database operation that undoes any changes made to the database during an incomplete transaction. If a transaction is not committed and the database or system shuts down, the changes made by the transaction may be lost or incomplete. In this case, the database management system (DBMS) will automatically perform a rollback operation to restore the database to its previous state before the transaction began.

The rollback process ensures that the database remains consistent and that any incomplete or inconsistent data is removed. This is an important feature of database management systems, which ensures the integrity and reliability of the data stored in the database.

Flashback is a database feature that allows a database administrator to view and recover historical data from the database. It does not automatically handle any incomplete transactions or shutdowns.

Commit is a database operation that finalizes a transaction and makes its changes permanent in the database. If a transaction is not committed, its changes are not permanent and can be rolled back.

View is a database object that is used to retrieve data from one or more tables in a specific format. It is not related to handling incomplete transactions or shutdowns.

11. The mathematical terms relation and tuple are referred to as?

- a. Table and key
- b. Table and Row
- c. Table and Column
- d. Table and Field

The correct answer is b. Table and Row.

In the context of databases, a relation refers to a table that contains data organized in rows and columns. Each row in the table represents a unique record or entity, while each column represents a specific attribute or field of the record.

A tuple is another term for a row in a database table. It refers to a single instance of a record or entity, which consists of a set of attribute values. In relational database theory, a tuple is an ordered set of values that uniquely identifies a record in a table.

Therefore, option b. Table and Row are the correct terms for relation and tuple, respectively, in the context of databases. Option a. Table and Key, option c. Table and Column, and option d. Table and Field are also important concepts in database management, but they are not related to the terms relation and tuple.

12. During installing a new video card into a desktop computer, what type of expansion slot is preferred today for high-performance graphics adapters?

- a. AGP
- b. PCI
- c. SATA
- d. PCIe

The correct answer is d. PCIe (Peripheral Component Interconnect Express).

PCIe is the preferred expansion slot for high-performance graphics adapters in modern desktop computers. PCIe is a high-speed serial bus interface that provides faster data transfer rates and improved performance compared to its predecessors, such as AGP (Accelerated Graphics Port) and PCI (Peripheral Component Interconnect).

PCIe comes in several versions, including PCIe 1.0, PCIe 2.0, PCIe 3.0, PCIe 4.0, and PCIe 5.0. Each version provides increased bandwidth and performance compared to the previous version, with PCIe 5.0 being the latest and fastest version available as of 2021.

AGP was a dedicated expansion slot for graphics cards that was popular in the late 1990s and early 2000s. It provided faster data transfer rates than the standard PCI slot, but it has been largely replaced by PCIe in modern computers.

PCI is a standard expansion slot that can be used for various types of expansion cards, including graphics cards. However, its lower bandwidth and slower data transfer rates make it less suitable for high-performance graphics applications.

SATA (Serial Advanced Technology Attachment) is a type of interface used for connecting storage devices, such as hard disk drives and solid-state drives, to a computer's motherboard. It is not an expansion slot and is not used for connecting graphics cards.

13. Digital signatures are primarily designed to provide additional protection with electronic messages in order to ensure

- a. Sender verification
- b. Message read by unauthorized party
- c. Message deletion
- d. Message modification

The correct answer is **d**. Message modification.

Digital signatures are used to provide additional protection to electronic messages by ensuring that the message has not been modified or tampered with during transmission. A digital signature is created using a mathematical algorithm that generates a unique code that is associated with the message. The code is created using the sender's private key and can be verified using the sender's public key.

When the recipient receives the message, they can use the sender's public key to verify the digital signature and ensure that the message has not been modified or tampered with during transmission. If the digital signature is valid, the recipient can be confident that the message is authentic and has not been altered in any way.

Therefore, option d. Message modification is the correct answer. Digital signatures are not designed to verify the sender's identity (option a), prevent unauthorized reading of the message (option b), or prevent message deletion (option c). However, digital signatures can provide some level of assurance about the sender's identity, as the private key used to create the signature is unique to the sender.

14. is a tool that are applicable to object-oriented development and we often used to examine class variables associated with an object and, as a stress tester, which stress the capacity of the software

- a. Configuration management
- b. Version control
- c. Incremental compiler
- d. Debugger

The correct answer is d. Debugger.

A debugger is a tool that is commonly used in object-oriented development to examine class variables associated with an object and to test the capacity of the software. Debugging is the process of finding and fixing bugs or errors in software code.

In object-oriented programming, objects are instances of classes, and each object has its own set of variables or attributes that are associated with the class. A debugger can be used to examine these variables and to monitor their values during program execution. It can also be used to simulate different scenarios and test the capacity of the software to handle different types of inputs or stress conditions.

Configuration management and version control are software engineering practices that are used to manage changes to software code and to ensure that different versions of the code are properly documented and maintained.

An incremental compiler is a type of compiler that compiles only the parts of the code that have been modified or changed since the last compilation. This can help to speed up the compilation process and reduce the time required to build and test software applications. However, incremental compilers are not specific to object-oriented development and are used in a variety of programming languages and software development environments.

15. Identify the necessary tool for creating and testing your android apps on different virtual devices.

- a. A system image for the Android emulator
- b. Android SDK
- c. Android SDK tools and platform tools
- d. IntelliJ IDEA

The correct answer is a. a system image for the Android emulator.

The Android emulator is a virtual device that allows developers to test their apps on a variety of virtual devices with different configurations. In order to create and test apps on the Android emulator, developers need to have a system image installed for the version of Android they want to test.

Option b, Android SDK (Software Development Kit), is a collection of tools and resources that developers use to create Android apps. The Android SDK includes the Android emulator, but it does not include the system images needed to run the emulator.

Option c, Android SDK tools and platform tools, refers to the command-line tools and utilities that are included with the Android SDK. These tools are used for various tasks such as building and debugging Android apps, but they do not include the system images needed to run the emulator.

Option d, IntelliJ IDEA, is an integrated development environment (IDE) that is commonly used by Android developers for creating and testing Android apps. While IntelliJ IDEA includes support for creating and testing Android apps, it does not include the system images needed to run the emulator. The system images need to be downloaded separately through the Android SDK Manager.

16. What is the importance of ssh command?

- a. performed in real-time
- b. performed in a synchronous way
- c. to restrict actions available to users
- d. to connect remote hosts via an authenticated and encrypted channel.

The correct answer is d. to connect remote hosts via an authenticated and encrypted channel.

SSH is a cryptographic network protocol that is used to secure network communications between two computers, such as connecting to a remote server from a local computer. The SSH command is used to initiate an SSH session between the local computer and the remote server, allowing users to access the remote server's command-line interface and execute commands remotely.

The SSH protocol provides a secure and encrypted communication channel between the local computer and the remote server, which helps to prevent eavesdropping, tampering, and other types of attacks that can compromise the security and privacy of the communication.

Some of the benefits and importance of using the SSH command for remote access include:

Authentication: SSH provides strong authentication mechanisms to verify the identity of the remote server and the user, which helps to prevent unauthorized access and protect against phishing attacks.

Encryption: SSH encrypts all communications between the local computer and the remote server, which helps to protect the confidentiality and integrity of the data being transmitted.

Remote access: SSH allows users to access and manage remote servers from anywhere in the world, which can be useful for system administration, remote development, and other tasks.

Therefore, the importance of the SSH command lies in its ability to establish a secure and encrypted channel for remote access and communication.

17. Which of the following programming style employs Invariant relationships abstractions?

- a. Logic-oriented
- b. Constraint-oriented
- C. Object-oriented
- d. Procedure-oriented

The correct answer is c. Object-oriented.

Invariant relationships are a fundamental concept in object-oriented programming (OOP) that help ensure the correctness and consistency of an object's state. An invariant is a condition that must always be true for an object, regardless of its current state or the operations performed on it.

In OOP, objects are modeled as entities that have state, behavior, and identity. Invariant relationships are used to define the constraints or rules that govern an object's state, and they help to ensure that the object remains in a valid and consistent state throughout its lifecycle.

For example, consider a bank account object in an OOP program. An invariant relationship for this object might be that the account balance must always be greater than or equal to zero. This invariant can be enforced using encapsulation, which is a key principle of OOP. The account balance would be stored as a private variable within the object, and the object's methods would be designed to ensure that the balance is never set to a negative value.

Therefore, the programming style that employs invariant relationships abstractions is object-oriented programming (OOP), which is option c. Logic-oriented programming (option a), constraint-oriented programming (option b), and procedure-oriented programming (option d) may also use abstraction and other programming concepts, but they are not specifically focused on invariant relationships as a core concept.

18. Connectionless communication that provides no error recovery and no delivery guarantee.

- a. Simple mail transfer protocol
- b. File transfer protocol
- c. Internet Protocol
- d. User Datagram Protocol

The correct answer is d. User Datagram Protocol (UDP).

UDP is a connectionless transport layer protocol that is used for sending datagrams (packets) over a network. Unlike TCP (Transmission Control Protocol), which is a connection-oriented protocol that provides reliable delivery of data, UDP provides no error recovery and no delivery guarantee. This makes it a simpler and faster protocol than TCP, but also less reliable.

UDP is often used for applications that require low latency and high-speed data transfer, such as online gaming, video streaming, and VoIP (Voice over Internet Protocol). These types of applications can tolerate some loss of data or occasional errors, and they prioritize speed and responsiveness over reliability.

Option a, Simple Mail Transfer Protocol (SMTP), is a protocol used for sending and receiving email messages. SMTP is a connection-oriented protocol that provides reliable delivery of email messages.

Option b, File Transfer Protocol (FTP), is a protocol used for transferring files over a network. FTP can use either TCP or UDP as its transport layer protocol, but it typically uses TCP for its reliability and error recovery features.

Option c, Internet Protocol (IP), is a network layer protocol that provides the basic routing and addressing functions for data packets on the Internet. IP can use either TCP or UDP as its transport layer protocol, depending on the specific application and requirements.

19. To support the 802.1Q protocol, a trunk interface needs to be configured. What command will accomplish this?

- a. Switch(config-if)# switchport 802.1q
- b. Switch(config-if)# switchport mode trunk 802.1q
- c. Switch(config-if)# switchport encapsulation trunk 802.1q
- d. Switch(config-if)# switchport trunk encapsulation 802.1q

The correct answer is c. Switch(config-if)# switchport encapsulation trunk 802.1q.

To support the 802.1Q protocol, which is used for VLAN (Virtual Local Area Network) tagging, a trunk interface needs to be configured on a network switch.

A trunk interface is a type of Ethernet interface that carries multiple VLANs over a single physical link.

To configure a trunk interface on a Cisco network switch, the "switchport encapsulation" command is used to specify the encapsulation method for the trunk.

The correct command to configure a trunk interface with 802.1Q encapsulation is:

```
Switch(config-if)# switchport encapsulation trunk 802.1q
```

Option a, "switchport 802.1q", is not a valid command to configure a trunk interface with 802.1Q encapsulation.

Option b, "switchport mode trunk 802.1q", is a command to configure a switchport to operate in trunk mode, but it does not specify the encapsulation method.

Option d, "switchport trunk encapsulation 802.1q", is a valid command to configure a trunk interface with 802.1Q encapsulation, but the correct syntax for this command is:

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

Overall, the correct command to configure a trunk interface with 802.1Q encapsulation on a Cisco network switch is

```
Switch(config-if)# switchport encapsulation trunk 802.1q.
```

20. Referencing the following HTML listing, how would you style only the first paragraph inside the footer element to have a smaller font size?

- a. footer p.first-child { font-size: x-small; }
- b. footer->p,first-child {font- size: x-small;}
- c. footer p:first-child { font-size: x-small; }
- d. footer:p:first-child { font-size: x-small;}

The correct answer is c. footer p:first-child { font-size: x-small; }.

To style only the first paragraph inside the footer element with a smaller font size in HTML, we can use CSS (Cascading Style Sheets) to apply the styling. In this case, we want to select the first child paragraph element (i.e., the first paragraph element) that is a descendant of the footer element.

The correct CSS selector to accomplish this is:

```
footer p:first-child { font-size: x-small; }
```

This selector targets the first child paragraph element that is a descendant of a footer element, and applies the font-size property with a value of x-small to it.

Option a, "footer p.first-child { font-size: x-small; }", is incorrect because the ".first-child" selector selects elements that are the first child of their parent, but it does not select the first occurrence of an element type within its parent.

Option b, "footer->p,first-child {font-size: x-small;}", is incorrect because "->" is not a valid selector in CSS, and the comma (,) separates two independent selectors.

Option d, "footer:p:first-child { font-size: x-small;}", is incorrect because it would select the first occurrence of a paragraph element that is also the first child of its parent, but it would not select only the first paragraph element that is a descendant of the footer element. The space between "footer" and "p" indicates that any paragraph element that is a descendant of the footer element would be selected.

21. is HTTP status code for client error such as page not found.

a. 5xx

b. 4xx

c. 3xx

d. 1xx

The correct answer is b. 4xx.

HTTP status codes are 3-digit codes that are returned by web servers to indicate the status of a client's request. The first digit of the status code indicates the general category of the response, and there are 5 categories:

1xx (Informational): The request was received, and the server is continuing to process it.

2xx (Successful): The request was successfully received, understood, and accepted.

3xx (Redirection): The request requires further action to complete, such as following a redirect.

4xx (Client Error): The request contains bad syntax or cannot be fulfilled by the server, such as a page not found error (404).

5xx (Server Error): The server encountered an error while processing the request.

Therefore, the HTTP status code for a client error such as a page not found error is in the 4xx category. Specifically, the 404 status code is used to indicate that the requested resource (such as a web page) could not be found on the server.

Option a, 5xx, is used to indicate server errors, such as when the server is unable to fulfill the request due to an internal error.

Option c, 3xx, is used for redirection, such as when a requested resource has been moved to a different URL and the client should redirect to the new URL.

Option d, 1xx, is used for informational status codes, such as when the server is continuing to process the client's request.

22. The correct order of query optimization is

- a. Querying, Validating ,Scanning, Parsing
- b. Querying, Parsing, Validating, Scanning
- c. Querying, Scanning, Parsing, Validating
- d. Querying, Scanning,, Validating, Parsing

The correct order of query optimization is: b. Querying, Parsing, Validating, Scanning.

Query optimization is the process of improving the performance and efficiency of database queries by selecting the most efficient query execution plan. The following is the order of the steps involved in query optimization:

Parsing: This is the process of breaking down the SQL query into its individual components and checking the syntax of the query for errors.

Validating: This step involves checking the query for semantic errors, such as checking that all referenced tables and columns exist, and that the user has the necessary permissions to access them.

Querying: This is where the database query optimizer selects the most efficient execution plan for the query, based on factors such as the size of the tables being queried, the available indexes, and the query complexity.

Scanning: This is where the database engine retrieves the necessary data from the tables, using the execution plan determined by the query optimizer.

Therefore, the correct order of query optimization is: Querying, Parsing, Validating, Scanning. Option b is the correct order as it follows this sequence.

23. Which function is operated in layer 2 (function of a switch)?

- a. Learning the MAC address by examining the destination MAC addresses
- b. Determining the forwarding interfaces based upon the destination MAC address and tables
- c. Repeating the electrical signal to all ports

d. Forwarding the data based upon logical addressing

The correct answer is b. Determining the forwarding interfaces based upon the destination MAC address and tables.

A switch is a networking device that operates at layer 2 of the OSI (Open Systems Interconnection) model, which is the Data Link layer. The Data Link layer is responsible for the reliable transfer of data between adjacent network nodes and includes two sub-layers: the Logical Link Control (LLC) sub-layer and the Media Access Control (MAC) sub-layer.

The main functions of a switch at layer 2 are:

- a. Learning the MAC address by examining the source MAC addresses of incoming frames.
- b. Determining the forwarding interfaces based upon the destination MAC address and tables that map MAC addresses to interfaces.
- c. Filtering frames based on the destination MAC address to reduce unnecessary traffic on the network.
- d. Forwarding frames to the appropriate interface based on the destination MAC address.

The correct answer is **b.** determining the forwarding interfaces based upon the destination MAC address and tables that map MAC addresses to interfaces.

Out of these functions, option b - determining the forwarding interfaces based upon the destination MAC address and tables is the function that is operated in layer 2 of a switch. When

a switch receives a frame, it examines the destination MAC address in the frame header and looks up the MAC address in its forwarding table to determine which interface to forward the frame out of.

Option a - learning the MAC address by examining the source MAC addresses of incoming frames - is also a function of a switch at layer 2, but it is not the function that is being asked for in this question.

Option c - repeating the electrical signal to all ports - is a function of a hub, which operates at layer 1 of the OSI model.

Option d - forwarding the data based upon logical addressing - is a function of a router, which operates at layer 3 (Network layer) of the OSI model.

24. describe when an activity will start interacting with the user.

a. onCreate

b. onDestroy

c. onStop

d. onResume

The correct answer is d. onResume.

In Android app development, an Activity is a single screen with a user interface that the user can interact with. An Activity can go through several states during its lifecycle, such as when it is

created, started, resumed, paused, stopped, and destroyed.

The `onResume()` method is a callback method that is called when an Activity is brought to the foreground and becomes the active Activity that the user is interacting with. Specifically, `onResume()` is called after the Activity has been paused (either by the system or by another Activity) and is about to be resumed.

The `onResume()` method is a good place to perform any actions that need to be done when the Activity is being resumed, such as updating the user interface, refreshing data, or re-establishing connections to services or resources. It is also a good place to register any necessary listeners or receivers for handling user interactions.

Option a, `onCreate()`, is called when the Activity is first created, and is used for initializing the Activity and setting up the user interface.

Option b, `onDestroy()`, is called when the Activity is being destroyed and is used for releasing any resources or performing any cleanup tasks.

Option c, `onStop()`, is called when the Activity is no longer visible to the user and is used for releasing any resources that are no longer needed.

Overall, `onResume()` is the correct method to use when an Activity will start interacting with the user.

25. Which one of the following directories contains the configuration files?

a. `/etc/`

b. /dev/

c. /root/

d. /bin/

The correct answer is a. /etc/.

In Linux and Unix-based operating systems, the /etc/ directory contains configuration files and directories that are used by the system and applications. The name "etc" stands for "et cetera", meaning "and other things", indicating that the directory contains miscellaneous system files and configurations.

Some examples of the types of files that might be found in the /etc/ directory include system configuration files, network configuration files, user account information, and service configuration files.

Option b, /dev/, is a directory that contains device files that represent physical and virtual devices attached to the system.

Option c, /root/, is the home directory of the root user, which contains that user's personal files and configurations.

Option d, /bin/, is a directory that contains essential binary executable files that are required for the system to function, such as basic system utilities and commands.

Overall, the correct answer is option a, /etc/, as it is the directory that typically contains configuration files in Linux and Unix-based operating systems.

26. The super global array variables are accessible

- a. Anywhere
- b. Only inside functions
- c. Anywhere except in classes
- d. Only outside functions

The correct answer is a. Anywhere.

In PHP, the superglobal arrays are special predefined variables that can be accessed from anywhere in a script, including inside functions and classes. Superglobal arrays are available in every scope of a PHP script, including the global scope, function scope, and class scope.

There are several superglobal arrays in PHP, including:

`$_SERVER`: Contains information about server and execution environment

`$_GET`: Contains variables passed to the current script via the HTTP GET method

`$_POST`: Contains variables passed to the current script via the HTTP POST method

`$_FILES`: Contains information about uploaded files

`$_COOKIE`: Contains variables passed to the current script via HTTP cookies

`$_SESSION`: Contains variables that are available to the current session

Superglobal variables are always available and can be accessed from anywhere in the script, regardless of the scope. However, it is recommended to use them with caution and to properly sanitize and validate any user input before using it, in order to prevent security vulnerabilities such as injection attacks.

Therefore, the correct answer to the question is option a, anywhere, because superglobal arrays are accessible from any scope in a PHP script.

27. Is an insertion operator which is used for overloading.

a. Overloading <<

b. Overloading ||

c. Overloading +=

d. Overloading &&

The correct answer is a. Overloading <<.

In C++ programming, the insertion operator << is used for outputting values to the console or other output streams. However, the insertion operator can also be overloaded, which allows it to be used with custom data types and classes.

Overloading the insertion operator involves defining a function that takes an output stream object (such as `std::cout`) and a custom object as arguments, and returns the output stream object. This allows the custom object to be output in a custom format, using the insertion operator.

For example, consider a custom class named "Person" with member variables for the person's name and age. We can overload the insertion operator to output a Person object in the format "Name: [name], Age: [age]".

arduino

```
class Person {
```

```
public:
```

```
    std::string name;
```

```
    int age;
```

```
};
```

```
std::ostream& operator<<(std::ostream& os, const Person& person) {
```

```
    os << "Name: " << person.name << ", Age: " << person.age;
```

```
    return os;
```

```
}
```

With this overloaded operator, we can now output a Person object to the console using the insertion operator:

```
Person p;
```

```
p.name = "John";
```

```
p.age = 30;
```

```
std::cout << p << std::endl; // outputs "Name: John, Age: 30"
```

Therefore, the correct answer is option a, Overloading <<, as it is the insertion operator that is commonly used for overloading in C++ programming.

28. If you don't specify a return value, Android function will return_

a. False

b. True

c. Void

d. Unit

The correct answer is c. Void.

In Android app development, a function can be defined to perform a specific task or operation. A function can take parameters as input and return a value as output. The return type of a function specifies the type of value that the function will return.

If a function does not need to return a value, we can specify the return type as void. The void return type indicates that the function does not return any value.

For example, consider the following function that displays a message on the screen:

```
public void showMessage(String message) {  
    Toast.makeText(this, message, Toast.LENGTH_SHORT).show();  
}
```

In this function, the return type is void, indicating that it does not return any value. The function takes a String parameter named "message", which is used to display a message on the screen using a Toast.

If a function does not specify a return value and does not have a return statement, the function will still execute normally, but it will not return any value. The function will simply perform its intended task and then complete without returning anything.

Therefore, the correct answer is c. Void, as a function in Android that does not specify a return value will typically use void as its return type.

29. Which of the following operators is utilized in Java to allocate memory to an array variable?

- a. alloc
- b. new
- c. malloc
- d. new malloc

The correct answer is b. new.

In Java, the new operator is used to allocate memory to an array variable. The new operator is used to create objects, and when applied to an array variable, it allocates memory for the array and assigns a reference to the array to the variable.

For example, to create an array of integers with five elements, we can use the following code:

```
int[] numbers = new int[5];
```

This allocates memory for an array of five integers and assigns a reference to the array to the variable "numbers".

The "alloc" keyword is not used for allocating memory to an array variable in Java. It is not a valid keyword in the Java language.

The "malloc" function is a C/C++ library function that is used to allocate memory dynamically. It is not used in Java, as Java manages memory automatically through a process called garbage collection.

Therefore, the correct answer is b. new, as it is the operator used in Java to allocate memory to an array variable.

Context cleared

30. Identify the function that changes the output of `intSalary InputBox("What is your salary")` into numerical representation

a. `Convert()`

b. `Val()`

c. `Parse()`

d. `Int ()`

The correct answer is b. `Val()`.

The function that changes the output of `InputBox("What is your salary")` into numerical representation is the `Val()` function.

The `Val()` function takes a string as input and returns the numeric value contained in that string. Therefore, it can be used to convert the string input from the `InputBox()` function into a numerical representation that can be used for calculations or other purposes.

The other functions listed, `Convert()`, `Parse()`, and `Int()`, can also be used to convert strings to numerical values, but they have different syntax and behavior compared to `Val()`.

31. The statement " $x = 3 + 4^* 4$ " is executed. In this case what value will be assigned to the numeric variable x?

a. 92

b. 234

c. 19

d. 20

The correct answer is c. 19

The statement $x = 3 + 4^* 4$ will result in x being assigned a value of 19.

The order of operations in this statement is important. The exponentiation (^) operation has higher precedence than multiplication (*) and addition (+). Therefore, the expression $4^* 4$ is evaluated first, which is not a valid operation as the * operator requires an operand on both sides.

Assuming that the intended expression was 4^4 , which evaluates to 256, the corrected statement would be $x = 3 + 4^4$. Evaluating this expression would result in x being assigned a value of 259.

32. Which one of following will be the result of the below Java code?

a. 1

b. 1.5

O c. 6

O d. 0

NO Answer Because There is no source code

None of the given options is the correct answer for the below Java code as it contains a syntax error and will not compile:

```
int x = 5;  
  
double y = x / 3;  
  
System.out.println(y);
```

The issue is with the line `double y = x / 3;`. Here, `x` is an `int` and `3` is also an `int`, so the result of the division operation is an `int` as well. Since `int` cannot be directly assigned to `double`, this will result in a compilation error.

To fix this, we can either cast `x` to `double` before the division operation or use a `double` literal for the divisor:

```
int x = 5;  
  
double y = (double) x / 3;  
  
System.out.println(y);
```

or

```
int x = 5;  
  
double y = x / 3.0;  
  
System.out.println(y);
```

Both of these will result in `y` being assigned a value of `1.6666666666666667` (which will be printed as `1.6666666666666667` by the `println()` statement).

33. Which access/contention method is used for Ethernet?

- a. Token passing
- b. 802.2
- c. CSMA/CA
- d. CSMA/CD

The correct answer is d. CSMA/CD.

The access/contention method used for Ethernet is CSMA/CD (Carrier Sense Multiple Access with Collision Detection).

In CSMA/CD, before transmitting a frame over the Ethernet network, a device listens to the network to check if the medium is idle. If the medium is busy, the device waits until it is idle. Once the medium is idle, the device starts transmitting the frame. While transmitting, the device continues to listen to the network to detect any collisions (i.e., if two or more devices transmit at the same time, their signals will collide and become garbled). If a collision is detected, the device stops transmitting, sends a jam signal to indicate the collision, waits for a random amount of time, and then retries transmission.

CSMA/CD is used in Ethernet networks that use coaxial cable or twisted pair cables as the physical medium. It is not used in wireless Ethernet networks (which use CSMA/CA instead) or in token passing networks (such as Token Ring).

34. A Communication between applications and the Android Operating System is handled by

- a. Content Provider
- b. Broadcast Receivers
- c. Activities
- d. Service

Communication between applications and the Android Operating System is typically handled by using Android components such as Activities, Services, Broadcast Receivers, and Content Providers.

Out of the given options, the component that is primarily responsible for handling user interactions and managing the user interface of an application is the Activity. Activities are used to present a graphical user interface (GUI) to the user and interact with the user to perform some action.

Services, on the other hand, are used to perform long-running operations in the background without any user interaction. They can be used to perform tasks even when the application is not in the foreground or not running at all.

Broadcast Receivers are used to respond to system-wide broadcast events such as battery low, network connectivity change, etc.

Content Providers are used to manage access to a structured set of data (such as a database) and provide a mechanism for inter-process communication (IPC) between different applications.

Therefore, the correct answer to this question is none of the given options as it is incomplete. To handle communication between applications and the Android Operating System, a combination of these components is typically used based on the specific use case.

35. What is the correct way to open the file "time.txt" as readable?

- a. `open("time.txt","read");`
- b. `fopen("time.txt","r+");`
- c. `fopen("time.txt","r");`
- d. `open("time.txt");`

The correct answer is c. `fopen("time.txt","r");`

The "r" mode specifies that the file should be opened for reading. This mode assumes that the file already exists, and if the file does not exist, it will return a NULL pointer.

Option a is incorrect because the `open()` function is not used to open a file in C programming language. Instead, the `fopen()` function is used.

Option b is incorrect because the "r+" mode specifies that the file should be opened for both reading and writing. If the file does not exist, it will return a NULL pointer. This mode is not suitable for opening a file for reading only.

Option d is also incorrect because the `open()` function is not used to open a file in C programming language, and it does not specify the mode in which the file should be opened.

36. Tuples that can form a relation state r of R . The constraint is that, for any two tuples t_1 and t_2 in r that have $t_1[X] = t_2[X]$ they must also have $t_1[Y] = t_2[Y]$ Which normalization level describes this concept?

a. 2NF

b. BCNF

c. 3NF

d. 1NF

The correct answer is c: 3NF.

The concept described in the question is an example of functional dependency. Specifically, it is a transitive dependency between non-key attributes.

The normalization level that deals with functional dependencies is the Third Normal Form (3NF). 3NF requires that every non-key attribute in a relation is functionally dependent on the primary key, and there are no transitive dependencies between non-key attributes. In other words, if $A \rightarrow B$ and $B \rightarrow C$ are functional dependencies in a relation, then A must be a superkey or part of a candidate key, and B and C must be directly dependent on A .

In the given scenario, the constraint that "for any two tuples t_1 and t_2 in r that have $t_1[X] = t_2[X]$, they must also have $t_1[Y] = t_2[Y]$ " is a functional dependency between the non-key attributes X and Y . Therefore, to satisfy the 3NF requirement, we need to separate the attributes X and Y into a separate relation, where X is the primary key and Y is functionally dependent on X .

Therefore, the correct answer to this question is option C: 3NF.

37. Is threats of a database, which can occur due to creation, insertion, updating, changing the status of data, and deletion.

- a. Loss of confidentiality
- b. Loss of availability
- c. Loss of integrity
- d. Loss of accountability

The correct answer is: c. Loss of integrity

The threats of a database, which can occur due to creation, insertion, updating, changing the status of data, and deletion, are primarily related to the loss of data integrity.

Data integrity refers to the accuracy and consistency of data over its entire lifecycle. The threats to data integrity can occur due to various reasons such as human errors, software bugs, hardware failures, and malicious activities by unauthorized users. Examples of such threats include:

- Insertion of incorrect or invalid data into the database
- Updating or modifying data in an unauthorized manner
- Deletion or removal of critical data from the database
- Corruption of data due to hardware or software failures
- Unauthorized access to the database by malicious users

38. You install a new NIC for a user, and he asks "what it does?". Which of the following best characterizes the functions of this device?

- a. Input/Output
- b. Storage
- c. Multimedia
- d. Communications

The correct Answer is: c. Communications

The function of a NIC (Network Interface Card) is related to communications.

A NIC is a hardware component that is used to connect a computer to a network. It provides the physical interface between the computer and the network, allowing the computer to communicate with other devices on the network. The NIC typically provides a wired or wireless connection to the network and is responsible for transmitting and receiving data over the network.

39. Application-associated processing in the background is handled by

- a. Activities
- b. Broadcast Receivers

c. Service

d. Content Provider

The correct Answer is: c. service

Application-associated processing in the background is typically handled by using a Service component in Android.

A Service is an Android component that runs in the background to perform long-running operations without any user interaction. It can be used to perform tasks even when the application is not in the foreground or not running at all. A Service can run in the same process as the application or in a separate process.

Activities are used to handle user interactions and manage the user interface of an application. Broadcast Receivers are used to respond to system-wide broadcast events such as battery low, network connectivity change, etc. Content Providers are used to manage access to a structured set of data (such as a database) and provide a mechanism for inter-process communication (IPC) between different applications.

40. On a router, you have numerous routes set up. What command only displays static routes?

- a. Router# show ip routes static
- b. Router# show ip routes
- c. Router# show ip static routes
- d. Router# show static routes

The correct Answer is: a. Router# show ip routes static

The correct command to display only static routes on a router is:

```
Router# show ip route static
```

This command will display only the static routes that have been manually configured on the router. If there are no static routes configured, the command will not display any output.

41. one of the following is not disadvantage of simple file processing.

- a. concurrent access anomalies
- b. program-data independence
- c. data separation and isolation
- d. data redundancy and inconsistency

The correct answer is: d. data redundancy and inconsistency

Simple file processing can lead to data redundancy and inconsistency because data may be duplicated across multiple files, and there is no standardized way to update or delete data. This can lead to inconsistencies and errors in the data. Additionally, simple file processing does not provide any mechanisms for enforcing data integrity, such as constraints or data validation rules.

Concurrent access anomalies and data separation and isolation are also disadvantages of simple file processing. Concurrent access anomalies can occur when multiple users or programs try to access the same file simultaneously, leading to conflicts and errors. Data separation and isolation can also be a problem, as there is no way to enforce relationships between data in different files.

42. _ is the default file system type of linux

- a. ext2
- b. ext
- c. ext3
- d. mimx

The default file system type of Linux is ext4.

Among the given options, ext2, ext, and ext3 are also file system types used by Linux. However, ext4 is the default file system type for most Linux distributions since it offers improved performance, scalability, and reliability compared to its predecessors.

mimx is not a valid file system type used by Linux.

43. in order to establish a neighborship, which is need for EIGRP routers.

- a. matching delay metrics
- b. matching K metrics
- c. matching areas
- d. matching bandwidth metrics

The correct answer is: b. matching K metrics

In order to establish a neighborship, EIGRP routers need to have matching K values.

EIGRP (Enhanced Interior Gateway Routing Protocol) uses a composite metric to calculate the best path to a destination network. The composite metric takes into account several factors, including bandwidth, delay, reliability, load, and MTU (Maximum Transmission Unit). The K

value is a configurable parameter that determines the relative importance of each of these factors in the composite metric calculation.

When EIGRP routers exchange routing information, they compare their K values to determine if they are compatible. If the K values match, the routers can establish a neighborship and exchange routing information. If the K values do not match, the routers cannot establish a neighborship and will not exchange routing information.

Matching delay, bandwidth, and other metrics are also important for accurate routing calculations, but matching K values is the primary requirement for establishing an EIGRP neighborship.

44. what is the maximum data transfer rate of USB high speed?

- a. 10 Gbps
- b. 480 Mbps
- c. 5 Gbps
- d. 12 Mbps

Answer b. The maximum data transfer rate of USB High Speed is 480 Mbps.

USB High Speed, also known as USB 2.0, was introduced in 2000 and is still widely used today. It has a maximum signaling rate of 480 Mbps, which translates to a maximum data transfer rate of 60 MB/s. This is significantly faster than the original USB Full Speed specification, which had a maximum signaling rate of 12 Mbps and a maximum data transfer rate of 1.5 MB/s.

USB 3.0 and later versions offer even faster data transfer rates, with USB 3.1 Gen 2 offering a maximum signaling rate of 10 Gbps (or 1.25 GB/s) and USB 4.0 offering a maximum signaling rate of 40 Gbps (or 5 GB/s).

45. When you design a class diagram, which class hierarchies' relationship you are not going to apply?

- a. association
- b. application
- c. aggregation
- d. instantiation

Answer d. instantiation.

When designing a class diagram, the class hierarchy relationship of "instantiation" is not typically represented.

The four most common relationships in class diagrams are:

- a. Association: A relationship between two classes that indicates that objects of one class are connected to objects of another class. For example, a Car class might have an association with a Driver class to indicate that a car is associated with a driver.
- b. Aggregation: A relationship between two classes that indicates that one class is part of another class. For example, a Car class might have an aggregation relationship with a Wheel class to indicate that a car is made up of wheels.
- c. Composition: A stronger form of aggregation that indicates that one class owns another class. For example, a Car class might have a composition relationship with an Engine class to indicate that a car owns an engine.
- d. Inheritance: A relationship between two classes that indicates that one class is a specialized version of another class. For example, a SportsCar class might inherit from a Car class to indicate that a sports car is a specialized type of car.

Instantiation, on the other hand, is not a type of class hierarchy relationship. It simply refers to the process of creating an instance of a class, or an object, from a class definition.

46. one of the following cannot measure the quality of an abstraction?

- a. ease of use
- b. primitiveness
- c. sufficiency
- d. coupling

Answer d. Coupling cannot measure the quality of an abstraction.

Coupling refers to the degree to which one module or component depends on another module or component. It is a measure of the interdependence between different parts of a system. While coupling is an important factor to consider when designing a system, it is not directly related to the quality of an abstraction.

The three most commonly used metrics for measuring the quality of an abstraction are:

- a. Primitiveness: The degree to which an abstraction is composed of simple, basic elements that can be easily understood and combined to create more complex structures.
- b. Sufficiency: The degree to which an abstraction captures all the essential features and behaviors of the system being modeled, without including unnecessary or irrelevant details.
- c. Ease of Use: The degree to which an abstraction is easy to understand, learn, and use, for both developers and end users.

These metrics can be used to evaluate the quality of an abstraction and identify areas for improvement.

47. Which factor that measures the quality of the management process?

- a. Budget management
- B. Project team
- C. Project plan
- D. Results

Answer a. Budget management.

Budget management is a factor that measures the quality of the management process.

Effective budget management is an important aspect of project management, as it involves planning, allocating, and tracking resources to ensure that the project stays within its budget constraints. A well-managed budget can help to ensure that the project is completed on time and within scope, while also minimizing the risk of cost overruns or other financial issues.

Other factors that measure the quality of the management process include:

Project team: The quality and effectiveness of the project team can have a significant impact on the success of the project. A well-organized and motivated team can help to ensure that the project is completed on time and to a high standard.

Project plan: The quality of the project plan is also a key factor in the success of the project. A well-designed and comprehensive project plan can help to keep the project on track and ensure that all stakeholders are aligned on the project goals and objectives.

Results: Ultimately, the success of the project is measured by the results that are achieved. A well-managed project should deliver the desired outcomes and benefits, while also meeting the needs and expectations of stakeholders.

48. Suppose that the selector in a Select Case block is the string variable myVar. Which of the following Case clause is Invalid.

- a. Case 739
- b. Case "Adams"
- c. Case (myVar Substring(0, 1))
- d. Case myVar Length

Answer d. Case myVar Length

In a Select Case block, the Case clauses must be expressions that can be evaluated at compile-time, and they must be of the same data type as the selector. In this case, the selector is a string variable, so the Case clauses must also be string expressions.

Option a is a valid Case clause, as it is an integer expression that can be implicitly converted to a string.

Option b is a valid Case clause, as it is a string expression that matches the data type of the selector.

Option c is also a valid Case clause, as it is a string expression that is created by calling the Substring method on the myVar variable.

Option d is invalid because the Length property of a string is not a valid expression to use as a Case clause. Instead, you could use a string constant to specify the length, such as:

Case "5"

This would match any string that has a length of 5 characters.

49. In Public key encryption. A wants to send an encrypted message to B, which statement is true?

- a. A encrypts message using B's private key
- d. A encrypts message using he private key
- c. A encrypts message using B's public key
- d. A encrypts message using public key

Answer c. A encrypts message using B's public key

In Public Key Encryption, A wants to send an encrypted message to B, A encrypts the message using B's public key.

Public key encryption is a type of cryptographic system that uses a pair of keys: a public key and a private key. The public key is used to encrypt messages, while the private key is used to decrypt them. In a public key encryption system, anyone can use the public key to encrypt a message, but only the owner of the private key can decrypt it.

To send an encrypted message to B, A would encrypt the message using B's public key, which A can obtain from a public key repository or by exchanging keys with B. Once the message is encrypted, only B can decrypt it using his/her private key.

Option a is incorrect because A cannot encrypt the message using B's private key, as the private key is only known to B and should be kept secret. If A encrypts the message using B's private key, anyone with B's public key can decrypt the message, which defeats the purpose of encryption.

Option b is also incorrect because A should not use his/her own private key to encrypt the message. The private key should be kept secret and only used for decryption.

Option c is the correct answer because A should use B's public key to encrypt the message

Option d is not a valid answer because it is too vague. A needs to know whose public key to use for encryption.

50. Assume: Biruk wants to send a secure message M to Alem, and they want to assure its integrity and confidentiality, How do they make it possible if they use public crypto? (Consider public and Private key pair KB, and K-B respectively for Biruk, X-A and K-A respectively for Alem).

a. $K-A-(K-B(M))$

b. $K+B(K-A-(M))$

c. $K-B(K+A(M))$

d. $K-B(K-B(MD))$

The correct option is C: $K-B(K+A(M))$.

If Biruk and Alem want to assure the integrity and confidentiality of the message using public-key cryptography, they can use the following steps:

Biruk encrypts the message M using Alem's public key K-A, creating ciphertext C.

Copy

$$C = K-A(M)$$

Biruk signs the ciphertext C using his private key K-B, creating a digital signature D.

Copy

$$D = K-B(C)$$

Biruk sends both the ciphertext and the digital signature to Alem.

Alem receives the ciphertext C and decrypts it using his private key K-A, recovering the original message M.

Copy

$$M = K+A(C)$$

Alem then verifies the integrity of the message by verifying the digital signature using Biruk's public key K-B.

Copy

$$K+B(C) = D$$

If the verification is successful, Alem can be confident that the message was sent by Biruk and that it has not been tampered with in transit.

51. defined for a simple connectionless communication that provides no error recovery and no delivery guarantee

- a. User Datagram Protocol
- b. File transfer protocol
- b. Simple mail transfer protocol
- C. Internet Protocol.

Answer a. User Datagram Protocol

The protocol that is defined for a simple connectionless communication that provides no error recovery and no delivery guarantee is User Datagram Protocol (UDP).

UDP is a transport layer protocol that provides a minimal level of service for sending datagrams over an IP network. It is a connectionless protocol, which means that it does not establish a dedicated end-to-end connection before sending data. Instead, each datagram is handled independently by the network, and there is no guarantee that all datagrams will arrive at their destination or arrive in the correct order.

UDP is often used for applications that do not require the reliability or error recovery features of connection-oriented protocols like TCP. Examples of applications that use UDP include real-time video and audio streaming, online gaming, and DNS (Domain Name System) lookups.

File Transfer Protocol (FTP) and Simple Mail Transfer Protocol (SMTP) are application layer protocols that are used for transferring files and sending email messages, respectively. They are not connectionless protocols like UDP.

Internet Protocol (IP) is a network layer protocol that provides the basic routing and addressing functions for sending data packets over an IP network. While IP is a connectionless protocol, it does not provide the minimal level of service that UDP does.

52. What command alters the group owner of are?

- a. chgrp
- b. group
- c. cgrp
- d. grp

Answer a. chgrp

The command that alters the group owner of a file or directory is "chgrp".

The "chgrp" command stands for "change group" and is used to change the group ownership of a file or directory on a Unix or Linux system. The syntax for the "chgrp" command is as follows:

`chgrp [OPTIONS] GROUP FILE`

Where "GROUP" is the new group owner that you want to assign to the file or directory, and "FILE" is the name of the file or directory that you want to change the group owner for.

For example, to change the group owner of a file named "example.txt" to a group named "sales", you would use the following command:

```
chgrp sales example.txt
```

This would change the group ownership of the "example.txt" file to the "sales" group.

Option b, c, and d are not valid commands for changing the group owner of a file or directory in UNIX or Linux systems.

53. which one is a valid name for a variable?

- A. 4 Tutorials
- B. 14.Tutorials
- C. T4-Tutorials
- D. 14_Tutorials

Answer D. The valid name for a variable is D. 14_Tutorials.

In most programming languages, variable names must follow certain rules and conventions. Typically, variable names can only contain letters, numbers, and underscores, and they cannot start with a number. Spaces and special characters are not allowed in variable names.

Option A is not a valid name for a variable because it starts with a number.

Option B is also not a valid name for a variable because it contains a period, which is not allowed in most programming languages.

Option C is not a valid name for a variable because it contains a hyphen, which is not allowed in most programming languages.

Option D is a valid name for a variable because it contains only letters and numbers, and it starts with a letter. The underscore character is also commonly used in variable names to separate words and improve readability.

54. Which one of the following creates a connection-oriented connection that provides reliable end-to-end transfer and uses window mechanism for flow control?

- a. Transport layer
- b. Transmission Control Protocol
- C. Session Layer
- D. Internet Protocol

Answer b. Transmission Control Protocol

The Transmission Control Protocol (TCP) creates a connection-oriented connection that provides reliable end-to-end transfer and uses a window mechanism for flow control.

TCP is a transport layer protocol that provides a reliable, connection-oriented service for sending data over an IP network. It uses a three-way handshake to establish a connection between two hosts, and it provides error detection, retransmission of lost packets, and flow control using a sliding window mechanism. TCP ensures that data is delivered in the correct order and without errors, making it suitable for applications that require reliable data transfer, such as file transfers and email.

The session layer is responsible for managing communication sessions between applications. It does not provide the same level of reliability and flow control as TCP.

The internet protocol (IP) is a network layer protocol that provides the basic routing and addressing functions for sending data packets over an IP network. IP is a connectionless protocol and does not provide the reliable, connection-oriented service provided by TCP.

55. Eavesdropping and packet sniffing are considered to be attacks of

- a. Non-repudiation
- b. Confidentiality
- c. Authentication
- d. Integrity

Answer b. Confidentiality

Eavesdropping and packet sniffing are considered to be attacks on the confidentiality of data.

Confidentiality refers to the protection of information from unauthorized disclosure. Eavesdropping and packet sniffing are techniques used by attackers to intercept and read network traffic, which can compromise the confidentiality of sensitive information such as passwords, credit card numbers, and other personal or confidential data.

Non-repudiation is the ability to prove the origin and integrity of a message or transaction, and to prevent the sender from denying that they sent the message. Attacks on non-repudiation involve attempts to falsify or dispute the authenticity of a message or transaction.

Authentication is the process of verifying the identity of a user or system, to ensure that they are who they claim to be. Attacks on authentication involve attempts to impersonate a legitimate user or system, or to obtain their login credentials through phishing, social engineering, or other means.

Integrity refers to the protection of data from unauthorized modification or tampering. Attacks on integrity involve attempts to alter, delete, or inject false data into a message or transaction, which can compromise the accuracy and trustworthiness of the data.

What is the proper syntax when using a message dialog box?

- a. `MessageBox.Show Hi There, "Hi`
- b. `MessageBox.Show Hi There, Hi`
- c. `MessageBox. Show(Hi there, Hi)`
- d. `MessageBox.Show(Hi there, "Hi)`

Answer d. `MessageBox.Show(Hi there, "Hi)`

The proper syntax when using a message dialog box depends on the programming language and framework being used, but generally, the correct syntax is:

```
MessageBox.Show("Message", "Title")
```

Where "Message" is the text of the message to be displayed, and "Title" is the title of the message box.

Option a is missing quotation marks around the message and title, and the title itself is not a valid string.

Option b is missing quotation marks around the message, and the title is not a valid string.

Option c is missing quotation marks around both the message and title.

Option d is missing closing quotation mark for the message and the title is not a valid string.

Therefore, the correct syntax is option c:

```
MessageBox.Show("Hi there", "Hi")
```

This will display a message box with the message "Hi there" and the title "Hi".

56. a system is in a _ State if there exists a set of transactions such that every transaction in the set is waiting for another transaction in set.

- A. Ready
- B. Idle
- C. Deadlock
- D. Waiting

Answer C. Deadlock

A system is in a "Deadlock" state if there exists a set of transactions such that every transaction in the set is waiting for another transaction in the set.

Deadlock is a situation in which two or more transactions are blocked and unable to proceed because each is waiting for one or more of the others to complete. This can occur in a multi-user database system when multiple transactions try to access the same resources (e.g., data records or system locks) simultaneously, leading to a circular waiting pattern. In a deadlock state, the system is essentially stuck and cannot make any progress until the deadlock is resolved.

The term "Ready" usually refers to a state where a transaction is waiting for resources to become available so that it can execute.

The term "Idle" usually refers to a state where a system or process is not currently processing any tasks.

The term "Waiting" can refer to a variety of states in which a transaction or process is waiting for some event or resource to become available. However, it does not specifically refer to the situation where a set of transactions are deadlocked.

57. What command will stop all plain-text password displays in a router configuration files for unencrypted passwords?

- A. Router (config)# service password-encryption
- B. Router (config)# service-password-encryption
- C. Router (config)# service password encryption
- D. Router(config)# enable password secret

Answer A. Router (config)# service password-encryption

The command that will stop all plain-text password displays in a router configuration file for unencrypted passwords is:

```
Router(config)# service password-encryption
```

This command enables password encryption for all plain-text passwords in the router configuration file. When this command is enabled, the router automatically encrypts all plain-text passwords using a weak encryption algorithm, which makes it more difficult for someone to read the passwords in the configuration file.

Option A is the correct command with the correct syntax.

Option B has an incorrect syntax with a dash between "service" and "password".

Option C also has an incorrect syntax with spaces between "service", "password", and "encryption".

Option D is a command that sets the password for privileged access to the router, but it does not encrypt plain-text passwords in the configuration file.

58. _is implemented by combining methods and attribute into a class.

- A. Encapsulation
- B. Polymorphism
- C. Abstraction
- D. Inheritance

Answer A. Encapsulation

Encapsulation is implemented by combining methods and attributes into a class.

Encapsulation is one of the fundamental principles of object-oriented programming (OOP). It refers to the practice of combining data and functions into a single unit, called a class, and restricting access to the data so that it can only be accessed through the functions or methods that are defined within the class.

In encapsulation, the data and the methods that operate on the data are treated as a single entity, and the inner workings of the class are hidden from the outside world. This helps to ensure that the data is used and modified in a controlled and consistent way, and it prevents external code from interfering with the internal state of the object.

Polymorphism, abstraction, and inheritance are also important concepts in OOP, but they are not directly related to the implementation of a class. Polymorphism refers to the ability of objects of different classes to be treated as if they were of the same class, while abstraction refers to the process of generalizing a complex system by reducing its complexity and focusing on the essential features. Inheritance refers to the ability of a class to inherit properties and methods from a parent or ancestor class.

**59. What will be the output of the following statement? txt.box. Text
=FormatCurrency(1234.567).**

- A. 1,234.57

- B. \$1234.57
- C. 1234. 567
- D. \$1,234. 57

Answer D. \$1,234. 57

The output of the following statement: `txt.box.Text = FormatCurrency(1234.567)` will depend on the locale and currency settings of the system where the code is executed.

However, in most English-speaking locales, the output would be:

\$1,234.57

The `FormatCurrency` function formats a numeric value as a currency string using the currency symbol and decimal separator for the current system locale. The default currency symbol for most English-speaking locales is the dollar sign (\$), and the decimal separator is a period (.) or comma (,) depending on the locale.

60. Which one of the following statement is not true when you analyzing a Java application that produces an output?

- a. In the method header `public static void main(String[] args)`, the word `void` means that the `main` method is an empty method.
- b. in the method header `public static void main(String[] args)`, the word `static` is an access specifier
- C. In the method header `public static void main(String[] args)`, the contents between the parentheses, `String[] args`, represent the type of argument that can be passed to the `main` method;
- D. In the the method header `public static void main(String [] args)`, the word `public` means that a method is accessible and sable even though no objects of the class exist.

Option A: "In the method header `public static void main(String[] args)`, the word `void` means that the main method is an empty method" is not a true statement when analyzing a Java application that produces an output.

Answer a. In the method header `public static void main(String[] args)`, the word `void` means that the `main` method is an empty method.

The `void` keyword in the method header `public static void main(String[] args)` does not mean that the method is empty. Instead, it indicates that the method does not return any value. In Java, a method that does not return a value is declared with the `void` keyword in its signature.

The `main` method is the entry point of a Java application, and it takes an array of strings as its argument. The method signature must be declared as `public`, `static`, and `void`, and it must be named `main`.

Option B: "In the method header `public static void main(String[] args)`, the word `static` is an access specifier" is a true statement in Java. The `static` keyword in the method signature indicates that the method can be called without creating an instance of the class.

Option C: "In the method header `public static void main(String[] args)`, the contents between the parentheses, `String[] args`, represent the type of argument that can be passed to the `main` method" is a true statement in Java. The `String[] args` parameter allows command-line arguments to be passed to the `main` method.

Option D: "In the method header `public static void main(String[] args)`, the word `public` means that a method is accessible and usable even though no objects of the class exist" is a true statement in Java. The `public` keyword in the method signature indicates that the `main` method can be accessed from any other class.

61. Which one of the following statements correctly describes a feature of OOP Programming?

- a. Inheritance allows to treat all of an object's methods and data as a single entity

b. Encapsulation allows a class to be a subclass of a superclass and thereby inherits public and protected variables and methods of the superclass.

c. Objects are the characteristics of the defined an attribute; they are properties of the attributes.

d. a method a self-contained block of program code that carries out some action, similar to procedures in procedural program.

Answer b. Encapsulation allows a class to be a subclass of a superclass and thereby inherits public and protected variables and methods of the superclass.

Option B: "Encapsulation allows a class to be a subclass of a superclass and thereby inherit public and protected variables and methods of the superclass" is not a correct statement regarding the features of object-oriented programming.

Encapsulation refers to the process of combining data and functions into a single unit, called a class, and restricting access to the data so that it can only be accessed through the functions or methods that are defined within the class. It is not directly related to inheritance.

Option A: "Inheritance allows to treat all of an object's methods and data as a single entity" is not a completely accurate statement regarding inheritance. Inheritance allows a subclass to inherit methods and data members from a superclass, but it does not treat them as a single entity. The subclass still has its own methods and data members, in addition to the ones inherited from the superclass.

Option C: "Objects are the characteristics of the define an attribute; they are properties of the attributes" is not a correct statement regarding the features of object-oriented programming. Objects are instances of classes, and they have attributes (data members) and behaviors (methods).

Option D: "A method is a self-contained block of program code that carries out some action, similar to procedures in procedural programming" is a correct statement regarding the features of object-oriented programming. A method is a function that is associated with an object or class, and it can perform actions on the object's data members or other objects.

62. The two dimensions that you should use when evaluating project risks are.

a. negative and positive

b. probability and impact

- c. Source and responsibility
- d. Cost and Schedule

Answer b. probability and impact

The two dimensions that you should use when evaluating project risks are probability and impact.

Probability refers to the likelihood that a risk event will occur, while impact refers to the extent of the consequences if the risk event does occur. By evaluating risks in terms of their probability and impact, project managers can prioritize risks and focus their resources on those risks that have the highest potential impact on the project.

Option A: "Negative and positive" is not a correct statement regarding the dimensions used to evaluate project risks. While risks can have both negative and positive outcomes, this is not a helpful way to evaluate and prioritize risks.

Option C: "Source and responsibility" is not a correct statement regarding the dimensions used to evaluate project risks. While it is important to identify the source of a risk and assign responsibility for managing it, this is not the same as evaluating the probability and impact of the risk.

Option D: "Cost and Schedule" are important factors to consider when evaluating project risks, but they are not the dimensions used to evaluate risks. The probability and impact of a risk event can have an impact on the cost and schedule of a project, but they are not the same as evaluating risks in terms of cost and schedule.

63. You have just replaced a processor in a computer and now need to add a cooling mechanism. What should you use to the cooling system to the processor?

- a. Thermal paste
- b. superglue
- c. Heat sink
- d. Fan

Answer d. fan

A heat sink is a passive cooling component that is attached to the processor to dissipate heat. It is typically made of metal and has fins that increase the surface area for heat to dissipate. A fan is usually mounted on top of the heat sink to increase the airflow and improve cooling efficiency.

Option A: "Thermal paste" is used to improve the thermal conductivity between the processor and the heat sink. It is applied between the processor and the heat sink to fill the microscopic gaps and improve heat transfer.

Option B: "Superglue" is not a suitable cooling mechanism for a processor. It is a type of adhesive that is used to bond objects together and does not have any cooling properties.

Option C: "Heat sink" is a component that is used to cool a processor, but it requires a fan to improve the cooling efficiency. A heat sink alone cannot provide sufficient cooling to a processor, especially for high-performance processors.

64. Which of the following is true regarding the command `switchport mode dynamic auto`?

- a. The interface will become a trunk if requested on the neighboring port.
- b. The interface will remain an access link if the neighboring port is configured as a trunk
- c. The interface will remain an access link if the native VLAN is changed
- d. The interface will become a trunk if the neighboring port is configured the same

Answer b: "The interface will remain an access link if the neighboring port is configured as a trunk".

When an interface is configured with "`switchport mode dynamic auto`", it will attempt to negotiate with its neighboring interface to determine whether it should operate as an access link or a trunk link. If the neighboring interface is configured as a trunk, the local interface will

remain an access link and will not become a trunk. This is because the "dynamic auto" mode is passive and will only accept trunking if it is explicitly requested by the neighboring interface.

Option A: "The interface will become a trunk if requested on the neighboring port" is not a true statement regarding the "switchport mode dynamic auto" command. The "dynamic auto" mode is passive and will only accept trunking if it is explicitly requested by the neighboring interface.

Option C: "The interface will remain an access link if the native VLAN is changed" is not a true statement regarding the "switchport mode dynamic auto" command. The native VLAN is used for untagged traffic on a trunk link and does not affect whether an interface operates as an access link or a trunk link.

Option D: "The interface will become a trunk if the neighboring port is configured the same" is not a true statement regarding the "switchport mode dynamic auto" command. If both interfaces are configured with "switchport mode dynamic auto", they will both attempt to negotiate and will likely both remain as access links (unless the neighboring interface explicitly requests trunking).

The mathematical terms relation and tuple are referred to as?

- A. Table and Field
- B. Table and Row
- C. Table and Column
- D. Table and key

Answer B. table and row

In the context of relational databases, a relation is a table that stores data in rows and columns. Each row represents a single record in the table, and each column represents a specific attribute or field of the record.

A tuple is a single row or record in a relation or table. It contains a set of values that correspond to the columns or attributes in the table.

Option A: "Table and Field" refer to the relation and column or attribute of the relation, respectively.

Option C: "Table and Column" refer to the relation and column or attribute of the relation, respectively, but does not refer to the tuple or row.

Option D: "Table and key" refer to the relation and a specific column or set of columns that uniquely identify each row in the relation, respectively. While keys are an important concept in relational databases, they are not directly related to the terms relation and tuple.

65. In object-oriented development, through which diagrams we are not analyze the dynamic semantics of a problem or its implementation?

- a. Interaction diagram
- b. State transition diagrams
- c. Process diagrams
- d Script diagram

Answer c. Process diagrams.

Interaction diagrams (such as sequence diagrams and communication diagrams) and state transition diagrams are commonly used in object-oriented development to model the dynamic behavior of a system. These diagrams help to illustrate the interactions between objects and the changes in state that occur as the system executes.

Script diagrams are not a standard type of diagram used in object-oriented development. It is unclear what is meant by "script diagram" in this context, but it is not a common notation used for modeling the dynamic behavior of a system.

Process diagrams, on the other hand, are typically used in business process modeling and other types of process-oriented analysis. They may be used to model the flow of work or data through a system, but they do not provide a detailed view of the dynamic behavior of the system at the object or class level. As such, they are not typically used in object-oriented development to analyze the dynamic semantics of a problem or its implementation.

66. Which one of the following access methods used in a dedicated management channel?

- a. SSH
- b. In-band
- c. MAC
- d. Telnet

The access method used in a dedicated management channel is Out-of-band (Option not listed).

Out-of-band management refers to the use of a dedicated management channel that is separate from the main data channels used for the system. This allows administrators to manage the system and troubleshoot issues without interfering with regular system operations or risking security breaches.

Option A: "SSH" is an in-band access method that is used for secure remote access to a system over a network. It is typically used to access the command line interface of a system.

Option B: "In-band" access methods use the same communication channels as the main data channels of the system. This can be less secure and less reliable than out-of-band methods, as it can interfere with system operations and make it more difficult to troubleshoot issues.

Option C: "MAC" (Media Access Control) refers to the unique identifier assigned to each network interface controller (NIC) in a system. It is not an access method used for remote management.

Option D: "Telnet" is an in-band access method that is used for remote access to a system over a network. It is similar to SSH but is not as secure, as it does not encrypt data sent over the network. It is therefore not recommended for use in production environments.

67. If a method contains a local variable with the same name as one of its class's fields; the local variable field in that method's scope.

- a. Import
- b. shadows

- c. class
- d. collector

Answer b. shadows

If a method contains a local variable with the same name as one of its class's fields, the local variable shadows the class field in that method's scope (Option B).

This is known as variable shadowing, and it occurs when a variable declared within a certain scope has the same name as a variable declared in an outer scope. In the case of a method with a local variable that has the same name as a class field, the local variable takes precedence over the class field within the scope of the method.

For example, consider the following Java class:

```
public class Example {  
  
    private int x;  
  
    public void setX(int x) {  
  
        this.x = x; // "this.x" refers to the class field, while "x" refers to the local variable  
  
    }  
  
}
```

In this example, the class has a field called "x", and the "setX" method has a parameter also called "x". Within the scope of the "setX" method, the local variable "x" shadows the class field "x". To refer to the class field within the method, we use the "this" keyword to disambiguate the two variables.

Option A, C, and D are not correct as they do not describe the concept of variable shadowing.

68. Select the task involves when you write object-oriented programs.

- a. Creating a GUI environment for users is a natural use for object orientation
- b. Creating objects, which are blueprints for classes
- c. Creating classes, which are specific instances of objects

d. Creating applications that manipulate or use objects

The task that involves when you write object-oriented programs is creating objects, which are blueprints for classes (Option B) and creating applications that manipulate or use objects (Option D).

Object-oriented programming (OOP) involves creating classes, which define the blueprint or template for objects. Objects are instances of classes and represent specific instances of data or functionality. OOP is designed to model real-world objects or systems, and it provides a way to organize code into reusable, modular components.

Creating objects involves defining the attributes or properties of the object and its behavior or methods. This is typically done by instantiating a class and setting its properties or invoking its methods.

Creating applications that manipulate or use objects involves using objects to solve real-world problems or perform tasks. This may involve creating complex systems that involve multiple objects and interactions between them.

Option A is not a task that is necessarily involved in writing object-oriented programs. While creating a graphical user interface (GUI) may be a common use case for OOP, it is not a fundamental task involved in the OOP paradigm.

Option C is not accurate, as classes are not specific instances of objects. Rather, objects are instances of classes.

69. Assume the relation Staff(id: integer, ename: string, age integer, salary: real) Write SQL statement that increases employee psalary by 5%?

- a. `CHANGE Staff SET salary = salary + 0.05`
- b. `UPDATE Staff SET salary SET salary = salary1.05`
- c. `ALTER Staff salary = 0.05`

d. `MODIFY Staff SET salary = salary.05`

Answer b. `UPDATE Staff SET salary SET salary = salary1.05`

Explanation:

The SQL UPDATE statement is used to modify existing records in a table. In this case, we want to update the "Staff" table and increase the "salary" field by 5%. To do this, we multiply the current value of "salary" by 1.05, which effectively increases it by 5%.

Option A ("`CHANGE Staff SET salary = salary + 0.05`"), is not a valid SQL statement for increasing the salary by 5%. This statement would add a fixed value of 0.05 to the salary, rather than increasing it by a percentage.

Option C ("`ALTER Staff salary = 0.05`") is also not a valid SQL statement for increasing the salary by 5%. The ALTER statement is used to modify the structure of a table, such as adding or removing columns, but cannot be used to update existing records.

Option D ("`MODIFY Staff SET salary = salary*.05`") is also not a valid SQL statement for increasing the salary by 5%. This statement would multiply the salary by 0.05, which would result in a decrease in salary rather than an increase.

70. Which of the following assigns the value 3 to the 0th index of the temp array?

a. `3=temp[0];`

b. `temp[0]=3;`

c. `temp[3]=0;`

d. `temp(0)=3`

Answers b. temp[0] = 3;

Explanation:

Arrays in most programming languages are zero-indexed, meaning that the first element in the array has an index of 0. To assign a value to a specific element in the array, we use the square bracket notation followed by the index of the element we want to assign a value to.

In this case, we want to assign the value 3 to the 0th (first) element in the temp array. To do this, we use the assignment operator (=) to assign the value 3 to the element at index 0 of the array. The correct syntax for this is "temp[0] = 3;".

Option A ("3 = temp[0];") is not a valid syntax for assigning a value to an array element. This statement attempts to assign the value 3 to a constant value, which is not allowed.

Option C ("temp[3] = 0;") assigns the value 0 to the 3rd (fourth) element in the temp array, not the 0th element.

Option D ("temp(0) = 3;") is not the correct syntax for assigning a value to an element in an array. In most programming languages, the square bracket notation ([]) is used to access and assign values to array elements, not parentheses.

71. What is the correct HTML code for referring an external JavaScript?

- a. < script scr="myscript.js" type="text/javascript"> </script>
- b. < script>mystyle.css</script>
- c. <style scr="myscript.js"/> </style>
- d < script href="myscript.js" type="text/javascript"></script>

Answer a. `<script src="myscript.js" type="text/javascript"></script>`

Explanation:

To include an external JavaScript file in an HTML document, we use the `<script>` tag with the "src" attribute set to the URL of the JavaScript file. The "type" attribute specifies the MIME type of the script being included.

Option A ("`<script src="myscript.js" type="text/javascript"></script>`") correctly sets the "src" attribute to "myscript.js" and the "type" attribute to "text/javascript", making it the correct syntax for referring to an external JavaScript file.

Option B ("`<script>mystyle.css</script>`") is not valid syntax for including an external JavaScript file. The content between the script tags should be JavaScript code, not the name of a CSS file.

Option C ("`<style scr="myscript.js"/> </style>`") is not valid syntax for including an external JavaScript file. The "style" tag is used to define CSS styles within an HTML document, not to include external JavaScript files.

Option D ("`<script href="myscript.js" type="text/javascript"></script>`") is not valid syntax for including an external JavaScript file. The "href" attribute is used to specify the URL of a link in an HTML document, not to include external JavaScript files.

72. Which of the following is the correct tag to place an image on the right side of the window with the text filling the area to the left of the image?

- a. ``
- b. `<img src align="mountains.jpg" "right">`
- c. ``
- d. `<img ="mountainsjpg" src aligh="right">`

Option A is the correct answer. A. ``

Explanation:

To place an image on the right side of the window with the text filling the area to the left of the image, we can use the CSS float property. The float property specifies how an element should float in relation to the surrounding content.

To apply the float property to an image in HTML, we can use the `<img>` tag with the "style" attribute set to "float:right". This will cause the image to float to the right side of the window, with the text filling the area to the left of the image.

Option B ("`<img src align="mountains.jpg" "right">`") is not valid syntax for aligning an image to the right side of the window. The "align" attribute is used to specify how the text should be aligned in relation to an image, not how the image itself should be aligned.

Option C ("``") is not valid syntax for aligning an image to the right side of the window. The "align" attribute is used to specify how the text should be aligned in relation to an image, not how the image itself should be aligned. Placing "float:right" inside the "src" attribute is not valid syntax.

Option D ("`<img ="mountains.jpg" src aligh="right">`") is not valid syntax for aligning an image to the right side of the window. The syntax is incorrect, and the "align" attribute is misspelled as "aligh".

73. Among the four frames of organization _ addresses the question "how roles and responsibilities, coordination, and control are run?"

- a. Human resources
- b. Structure
- c. Symbolic
- d. Political

Answer b. Structure

Explanation:

The Structure frame is one of the four frames of organization described by Bolman and Deal in their book "Reframing Organizations". This frame focuses on the formal and informal structures within an organization and how they influence behavior, roles, responsibilities, and decision-making processes.

The Structure frame addresses questions such as:

What is the formal structure of the organization, and how does it influence behavior and decision-making?

What are the roles and responsibilities of different individuals and groups within the organization?

How does the organization coordinate its activities and resources across different departments or units?

How is control exercised within the organization, and what are the mechanisms for monitoring and enforcing performance?

Option A ("Human resources") is another frame of organization described by Bolman and Deal. The Human Resources frame emphasizes the importance of people within an organization and how they are managed, developed, and motivated.

Option C ("Symbolic") is another frame of organization described by Bolman and Deal. The Symbolic frame emphasizes the importance of culture, meaning, and symbolism within an organization and how they influence behavior and decision-making.

Option D ("Political") is another frame of organization described by Bolman and Deal. The Political frame emphasizes the importance of power, conflict, and coalitions within an organization and how they influence behavior and decision-making.

74. _____ IS a program that executes compiled. Java code on a specific platform.

- a. Java Compiler
- b. Java Virtual Machine
- c. Eclipse Editor
- d. Java Programming Manual

Answer b. Java Virtual Machine

Explanation:

Java is a high-level programming language that is compiled into bytecode, which can be executed on any platform that has a Java Virtual Machine (JVM) installed. The JVM provides a layer of abstraction between the compiled Java code and the underlying hardware and operating system, allowing Java programs to be executed on different platforms without requiring recompilation.

The JVM is responsible for interpreting the bytecode, managing memory, and providing security features such as sandboxing to prevent malicious code from accessing system resources. The JVM is available for many different platforms, including Windows, Linux, macOS, and mobile devices.

Option A ("Java Compiler") is a program that compiles Java source code into bytecode that can be executed on a JVM.

Option C ("Eclipse Editor") is a popular integrated development environment (IDE) for Java that provides a range of features for editing, debugging, and managing Java code.

Option D ("Java Programming Manual") is not a specific program or tool related to Java. It is likely referring to a manual or guidebook that provides information and guidance on Java programming.

75. In ER diagram Diamond/Ellipses represent.

- a. Relationship sets/Entity sets
- b. Entity sets/Attributes
- c. Attributes/Relationships sets
- d. Relationship sets/Attributes

Answer a. Relationship sets/Entity sets

Explanation:

ER diagrams (Entity-Relationship diagrams) are used to model and represent the relationships between entities in a database. They use various symbols to represent different components of the database, including entities, relationships, and attributes.

In an ER diagram, a diamond or ellipse symbol is used to represent a relationship set. A relationship set represents a collection of relationships between two or more entity sets. For example, in a database for a school, a relationship set might represent the relationship between the Students entity set and the Courses entity set, indicating which students are enrolled in which courses.

Entities are typically represented using rectangular boxes in an ER diagram, while attributes are represented using ovals or rounded rectangles.

76. _____ is the tag for the heading that is the largest and the most bold.

- a. H4
- b. H2
- c. H3
- d. H1

Answer d. H1

Explanation:

In HTML, the <h1> tag is used to define a heading of the highest level, which is typically the largest and most prominent heading on a webpage. The <h1> tag is often used for the main title or heading of a page, and it is typically styled to be bold and prominent.

The <h2>, <h3>, and <h4> tags are used for headings of progressively lower levels, with <h2> being the second highest level, <h3> being the third highest level, and so on. These tags are typically styled to be smaller and less prominent than the <h1> tag.

77. Which of the following assigns the value 3 to the 0th index of the temp array?

- a. 3=temp[0];
- b. temp[0]=3;
- c. temp[3]=0;
- d. temp(0)=3

Answer b. temp[0]=3;

Explanation:

In programming, arrays are used to store a collection of values of the same data type. Each value in an array is identified by an index, which is a numerical value that represents the position of the value in the array.

To assign a value to an element in an array, we use the name of the array followed by the index of the element in square brackets, and then the assignment operator (=) followed by the value we want to assign.

In this case, we want to assign the value 3 to the 0th index of the temp array. The correct syntax to do this is:

```
temp[0] = 3;
```

Option A ("3=temp[0];") is not valid syntax because the value being assigned must be on the left-hand side of the equals sign, while the variable or array element being assigned to must be on the right-hand side.

Option C ("temp[3]=0;") assigns the value 0 to the 3rd index of the temp array, not the 0th index.

Option D ("temp(0)=3") is not valid syntax because parentheses are not used to access elements in an array in most programming languages. Square brackets are used instead.

78. What programming language is used to create system calls in UNIX?

- a. C++
- b. Assembly Language
- c. Fortran
- d. C

Answer d. C

Explanation:

System calls are a fundamental part of the UNIX operating system, providing a way for programs to interact with the operating system kernel and access system resources such as files, processes, and network sockets.

In UNIX, system calls are typically implemented in the C programming language, which provides low-level access to system resources and the ability to interact with the operating system kernel through a standardized interface.

While it is possible to write system calls in other programming languages such as Assembly Language, C++ or Fortran, C is the most commonly used language for this purpose in UNIX.

79. When you develop a project management plan, which of the following ensures that the objectives of the project are achieved in the most appropriate way?

- a. Initiating
- b. Execution
- c. Controlling
- d. Planning

Answer d. Planning

Explanation:

The project management plan is a document that outlines the approach, processes, and procedures that will be used to accomplish the project objectives. The project management plan is developed during the planning phase of the project life cycle and is used as a guide throughout the project execution and control phases.

The planning process is where the project management plan is developed, and it involves defining the project scope, objectives, and deliverables, as well as identifying the tasks and activities that are required to complete the project. During this process, the project manager works with the project team to determine the most appropriate approach for achieving the project objectives, taking into account factors such as resources, timelines, budget, risks, and stakeholder expectations.

80. The object cannot be

- a. passed by value
- b. passed as function
- c. passed by reference
- d. passed by copy

Answer a. passed by value

Explanation:

In programming, passing an object as a parameter to a function or method can be done in several ways, including passing by value, passing by reference, and passing by copy.

When an object is passed by value, a copy of the object is created and passed to the function or method. This means that any changes made to the object within the function or method will not be reflected in the original object outside of the function or method.

In some cases, an object may be too large or complex to be efficiently passed by value. In these cases, it may be more appropriate to pass the object by reference or by copy.

81. What command is used to show and create files?

- a. lyrix
- b. ed
- c. vi

d. cat

Answer c.vi

Explanation:

Vi is a popular text editor in Unix and Linux operating systems that is used to create and edit text files. It is a powerful and versatile editor that can be used for a wide range of tasks, from simple text editing to programming and system administration tasks.

One of the key features of Vi is its ability to create and edit files. To create a new file in Vi, you can simply type "vi" followed by the name of the file you want to create. For example, to create a file called "myfile.txt", you would type:

```
vi myfile.txt
```

Once you have created a file in Vi, you can use various commands to show and edit the contents of the file. For example, you can use the "cat" command to display the contents of a file, or you can use various Vi commands to navigate, search, and edit the file as needed.

82. Assume: a computer lab where student demands the Administrator access to a Windows 11 system to install SQL Server: but that right should not be given to the student unless he/she is member of an employee. Which principle of cyber security is considered here?

- a. Separation of privileges
- b. Fail-safe
- c. Open design
- d. Least privileges

Answer d. least privileges

Explanation:

The principle of least privilege is a fundamental concept in cyber security that states that users, processes, and applications should only be given the minimum level of access and permissions necessary to perform their required tasks. This principle helps to reduce the risk of unauthorized access, data breaches, and other security incidents.

In the scenario described, the computer lab administrator is applying the principle of least privilege by only granting administrator access to the Windows 11 system to students who are also members of the employee group. This ensures that only authorized users who have a legitimate need for administrator access are granted that level of privilege, reducing the risk of unauthorized access, malware infections, and other security incidents.

The other principles of cyber security listed in the question are:

Separation of privileges: This principle involves dividing a system's functions and privileges among multiple users or processes to reduce the risk of unauthorized access or abuse of privileges.

Fail-safe: This principle involves designing a system or process to fail securely and minimize the impact of failures or errors.

Open design: This principle involves designing a system or process to be transparent and open to public scrutiny to improve security and identify potential vulnerabilities.

While these principles are also important in cyber security, they are not the primary principle being applied in the scenario described.

83. Which of the following is a physical memory format installed directly in today's desktop computer systems?

- a. HDD
- b. DIMM
- c. eMMC
- d. SSD

Answer b. DIMM

Explanation:

DIMM (Dual In-line Memory Module) is a physical memory format that is commonly used in modern desktop computer systems. DIMMs are small circuit boards that contain a series of memory chips and are designed to be installed directly into the motherboard of a computer.

DIMMs are a type of Random Access Memory (RAM) and are used to temporarily store data and program instructions that are currently in use by the computer. DIMMs are available in various capacities and speeds, and can be replaced or upgraded as needed to improve a computer's performance.

The other memory formats listed in the question are:

HDD (Hard Disk Drive): This is a type of non-volatile storage device that uses spinning disks and magnetic read/write heads to store and retrieve data.

eMMC (Embedded MultiMediaCard): This is a type of non-volatile storage device that is commonly used in mobile devices and other small form factor devices.

SSD (Solid State Drive): This is a type of non-volatile storage device that uses flash memory to store and retrieve data.

While these memory formats are also important in computer systems, they are not directly installed in the same way as DIMMs, and they serve a different purpose in the computer's overall architecture.

84. DES works by using

- a. exclusive ORing key bits with 64 bit blocks
- b. Permutation and substitution on 64 Bit blocks of plain text
- c. 4 rounds of substitution on 64 bit blocks with 56 bit keys
- d. only permutations on blocks of 128 bits

Answer b. Permutation and substitution on 64 Bit blocks of plain text

Explanation:

DES is a symmetric-key encryption algorithm that is used to encrypt and decrypt data. The algorithm works by using a 64-bit block of plain text and a 56-bit key to produce a 64-bit block of cipher text.

The encryption process in DES involves 16 rounds of permutation and substitution on the 64-bit block of plain text. During each round, the plaintext block is split into two 32-bit halves, which are then subjected to a series of bitwise operations, including permutation and substitution, using the 56-bit key.

The result of each round is then combined with the other half of the plaintext block, and the process is repeated for a total of 16 rounds. The final result is a 64-bit block of cipher text that is derived from the original plaintext block and the encryption key.

85. Which OSI reference model layer is responsible for translating data in a form that can be understood by the receiver?

- a. Application layer
- b. Presentation layer
- c. Transport layer
- d. Session layer

Answer b. Presentation layer

Explanation:

The OSI (Open Systems Interconnection) reference model is a conceptual framework that describes the communication functions of a telecommunication or computing system. The model is divided into seven layers, each of which is responsible for specific functions in the communication process.

The presentation layer is the sixth layer of the OSI model and is responsible for the formatting, encryption, and compression of data. This layer translates data into a form that can be understood by the receiver, regardless of the data representation used by the sender. The presentation layer also handles character encoding and is responsible for ensuring that the data is in the correct format for the application layer to use.

86. is a device that filters the information coming through an internet connection to a network or computer system

- a. Cookies
- b. Cyber safety
- c. Firewall
- d. Anti-Virus

Answer c. Firewall

Explanation:

A firewall is a network security device that monitors and filters incoming and outgoing network traffic based on predefined security rules. A firewall can be implemented using hardware, software, or a combination of both and is designed to protect a network or computer system from unauthorized access, malware, and other cyber threats.

A firewall typically operates by examining the data packets that are transmitted over the network and comparing them to a set of predefined rules. If a packet matches one of the rules, it is either allowed or blocked, depending on the specific rule. Firewalls can also be configured to log traffic and generate alerts when suspicious activity is detected.

Cookies are small data files that are stored on a user's computer by a website and are used to remember user preferences and track user behavior.

Cyber safety is a broad term that refers to the practices and measures used to protect computer systems, networks, and users from cyber threats.

An anti-virus is a software program that is designed to detect, prevent, and remove computer viruses and other malware.

87. During system design, what mechanisms/analysis is used to regulate how objects collaborate relative to the system architecture?

- a. Object diagrams
- b. Class diagrams
- c. Process diagrams
- d. Module diagrams

Answer b. Class diagrams

Explanation:

During system design, class diagrams are used to model the relationships and interactions between objects in a system. Class diagrams provide a graphical representation of the classes, interfaces, and their relationships that make up the system architecture.

Class diagrams are used to describe the static structure of the system, including the classes, attributes, and methods that comprise the system. They can also be used to model the dynamic behavior of the system, including the messages and interactions between objects during runtime.

Class diagrams are an important tool for regulating how objects collaborate relative to the system architecture because they provide a clear and concise representation of the objects and their relationships within the system. By analyzing the class diagram, system designers can ensure that the collaboration between objects is consistent with the system architecture and that the system functions as intended.

88. OSPF uses what multicast address for neighbor discovery?

- a. 224.0.0.5
- b. 224.0.0.4
- c. 224.0.0.6
- d. 224.0.0.7

Answer a. 224.0.0.5

Explanation:

OSPF is a link-state routing protocol that is used to determine the shortest path between two nodes on a network. OSPF uses multicast messages to discover and communicate with neighboring routers on the network.

In particular, OSPF uses the multicast address 224.0.0.5 for neighbor discovery. This multicast address is used by OSPF routers to send and receive Hello messages, which are used to establish and maintain neighbor adjacencies.

The other multicast addresses listed in the question are:

224.0.0.4: This multicast address is used by the Distance Vector Multicast Routing Protocol (DVMRP) to forward multicast traffic.

224.0.0.6: This multicast address is used by the OSPF protocol for Designated Router (DR) and Backup Designated Router (BDR) elections.

224.0.0.7: This multicast address is reserved for use by OSPFv3.

89. What needs to be configured according to the topology below, to allow traffic to be routed to the host if it enters RouterA with the destination address 198.44.4.212

- a. RouterA(config)#ip route 198.44.4.0 255.255.255.0 198.44.4.5
- B. RouterA(config)#ip route 198.44.4.0 255.255.255.0 fast 0/1
- C. Nothing needs to be done
- D. RouterA(config) #ip. route 198.44.4.0/24 fast 0/1

Answer a. RouterA(config)#ip route 198.44.4.0 255.255.255.0 198.44.4.5

Explanation:

In the given topology, the host with the IP address 198.44.4.212 is connected to Switch B, which is in turn connected to Router B. Router A is also connected to Router B via a serial link.

To allow traffic to be routed to the host with the destination address 198.44.4.212 if it enters RouterA, a static route needs to be configured on RouterA. The static route should specify the destination network (198.44.4.0), the subnet mask (255.255.255.0), and the next-hop router (198.44.4.6).

The correct command to configure the static route on RouterA is:

```
RouterA(config)#ip route 198.44.4.0 255.255.255.0 198.44.4.6
```

This command tells RouterA to forward any traffic with the destination IP address in the range of 198.44.4.0/24 to RouterB via the next-hop IP address of 198.44.4.6.

Option B is incorrect because it uses the interface name (FastEthernet0/1) instead of the next-hop IP address.

Option C is incorrect because a static route needs to be configured on RouterA to direct traffic to the destination host.

Option D is incorrect because it specifies the wrong format for the destination network (198.44.4.0/24) and uses the interface name (FastEthernet0/1) instead of the next-hop IP address.

90. Which information does the TCP header contain but the UDP header does not?

- a. Window
- b. Source port
- c Application layer data
- d. Checksum

The TCP (Transmission Control Protocol) header contains the information that the UDP (User Datagram Protocol) header does not, such as:

- a. Window
- d. Checksum

Options A and D are the correct answers.

Explanation:

TCP and UDP are transport layer protocols that are used to provide communication services between applications running on different hosts. Both protocols use headers to provide information about the communication session, including source and destination ports, sequence numbers, and other control information.

However, the TCP header contains additional information that the UDP header does not, including:

Window: The window field in the TCP header is used to manage flow control between the sender and receiver. It specifies the size of the receive window, which is the amount of data that the receiver is willing to accept at any given time.

Checksum: The checksum field in the TCP header is used to detect errors in the transmission of the data. It is calculated over the entire TCP segment, including the header and payload.

In contrast, the UDP header does not include a window field or a checksum field. Additionally, the UDP header does not include any information about the application layer data, unlike TCP which has a data offset field that specifies the size of the TCP header and the location of the application layer data.

91. Which of the following variable holds information about the web server and page

a. \$ SERVERO

b. \$_REQUESTD

c. \$SESSION]

d. \$_POST[]

None of the options provided is correct.

Explanation:

The variables listed in the options are PHP superglobal variables that are used to store information related to web requests and sessions:

\$SERVER: This superglobal variable holds information about the server and the execution environment, such as the server name, IP address, and request method.

\$_REQUEST: This superglobal variable is an associative array that contains the values of GET, POST, and COOKIE variables.

\$SESSION: This variable stores the session data for the current user.

`$_POST`: This superglobal variable is an associative array that contains the values of POST variables.

However, none of these variables directly holds information about the web server and page. To retrieve information about the web server and page, other variables and functions can be used, such as:

`$SERVER['SERVER_SOFTWARE']`: This variable holds the name and version of the web server software.

`$SERVER['REQUEST_URI']`: This variable holds the URI of the requested resource.

`$_SERVER['HTTP_USER_AGENT']`: This variable holds the user agent string of the client browser.

`get_headers()`: This function returns an array of headers sent by the server in response to an HTTP request.

92. HTTP response line with a status code that starts with 2xx means

- a. further action must be taken to complete the request
- b. success, action was successfully received, understood and accepted
- c. request received, processing continues
- d. server error occurred

Answer b. success, action was successfully received, understood and accepted

Explanation:

The first line of an HTTP response contains the status line, which includes the HTTP version, a status code, and a reason phrase. The status code is a three-digit number that indicates the status of the requested resource. The first digit of the status code indicates the general category of the response:

1xx: Informational - Request received, continuing process

2xx: Success - The action was successfully received, understood, and accepted

3xx: Redirection - Further action must be taken to complete the request

4xx: Client Error - The request contains bad syntax or cannot be fulfilled

5xx: Server Error - The server failed to fulfill an apparently valid request

Therefore, when the HTTP response line contains a status code that starts with 2xx, it means that the request was successful and the action was received, understood, and accepted. Examples of 2xx status codes include 200 OK, which indicates that the request was successful and the server has returned the requested data, and 204 No Content, which indicates that the request was successful but there is no data to return in the response.

93. When transaction T_i requests a data item currently held by T_j . T_i is allowed to wait only if it has a timestamp smaller than that of T_j (that is, T_i is older than T_j). Otherwise, T_i is rolled back (dies). This phenomenon is known as

- a. Wait-die
- b. Wound-wait
- c. Wait wound
- d. Wait

Answer a. Wait-die

Explanation:

Wait-die and wound-wait are two concurrency control protocols used to manage conflicts between transactions that are accessing the same data item. These protocols determine whether a transaction should be rolled back or allowed to wait for a conflicting transaction to release the required data item.

In the wait-die protocol, if a transaction T_i requests a data item currently held by T_j , T_i is allowed to wait only if it has a timestamp smaller than that of T_j (that is, T_i is older than T_j). If T_i 's timestamp is greater than or equal to T_j 's timestamp, T_i is rolled back (dies). This means that older transactions are allowed to wait for younger transactions, but younger transactions are not allowed to wait for older transactions.

In contrast, the wound-wait protocol allows a transaction T_i to wait for a data item held by T_j only if T_i 's timestamp is greater than or equal to T_j 's timestamp. If T_i 's timestamp is smaller than that of T_j , T_j is rolled back (wounded). This means that younger transactions are allowed to wait for older transactions, but older transactions are not allowed to wait for younger transactions.

94. The../notation in a relative path of hypertext reference matches.

- a. Search a folder
- b. Go down a folder
- c. Create a folder
- d. Open the parent folder

Answer d. Open the parent folder.

Explanation:

A hypertext reference (or hyperlink) is a link from one web page to another. Hyperlinks can be specified using absolute or relative URLs. A relative URL specifies the URL of the linked page relative to the current page.

The ../ notation in a relative URL is used to navigate up one level in the directory hierarchy. It represents the parent directory of the current directory. For example, if the current directory is /home/user/docs/ and a file in the directory /home/user/ needs to be linked, the relative URL would be ../file.html.

In contrast, the ./ notation in a relative URL represents the current directory and is used to specify a file or folder in the same directory as the current page. For example, if the current directory is /home/user/docs/ and a file in the same directory needs to be linked, the relative URL would be ./file.html.

95. Which tag used to add a background color for all <h1> elements?

- a. al.h1 (background-color:#FFFFFF)
- b. h1 (background-color:#FFFFFF)
- c.h1.all (background-color:#FFFFFF)
- d. All

Answer b. h1 {background-color:#FFFFFF;}

Explanation:

In HTML, CSS (Cascading Style Sheets) is used to style and layout web pages. CSS allows you to control the appearance of HTML elements by specifying properties such as color, font, and layout. To apply a style to an HTML element, you can use a CSS selector.

To add a background color for all <h1> elements, you can use the following CSS rule:

```
h1 {  
  
    background-color: #FFFFFF;  
  
}
```

This rule selects all <h1> elements and applies a background color of white (#FFFFFF). The curly braces {} define the block of styles to be applied to the selected elements.

Option A is incorrect because the selector al.h1 is not a valid CSS selector. The period . is used to select elements with a specific class, but al.h1 does not specify a valid class name.

Option C is incorrect because the selector h1.all is not a valid CSS selector. The period . is used to select elements with a specific class, but all is not a valid class name.

Option D is incorrect because it does not provide a valid CSS selector or rule.